

EUROPEAN DIGITAL DECADE STRATEGIC ROADMAP:

ESTONIA

2023 version

INTRODUCTION

The Estonian Digital Decade Strategic Roadmap is based on the current Estonian Digital Agenda 2030 (the national strategy for Estonian digital transformation), the Education strategy 2021–2035 and the Estonian Research and Development, Innovation and Entrepreneurship Strategy 2021—2035.

1. Estonian national roadmap comprises of the following:
 - a. Where applicable, the main planned, adopted and implemented policies, measures, and actions that contribute to achieving the general objectives and the digital targets.
 - b. Where available, national projected trajectories contributing to achieving the relevant digital targets set out in Article 4 of the Digital Decade Policy that are measurable at national level.
2. Where possible, this roadmap provides an estimate of the investments and resources needed to contribute to achieving the general objectives and the digital targets, as well as a general description of the sources of those investments, either private or public, including, where applicable, the planned use of Union programmes and instruments. The national roadmap includes Estonia's current intentions regarding multi-country projects to the extent these have been identified.

SECTION 1: ANALYSIS OF THE STATE OF PLAY OF DIGITAL TRANSFORMATION IN ESTONIA

STATE OF PLAY

According to the European Union Digital Decade Country Report 2023, Estonia is expected to make a positive contribution to the collective efforts to achieve the Digital Decade targets, namely for being at the forefront of the digitalisation of public services. However, further efforts are needed to ensure that the country's digital infrastructure is improved, especially the connectivity infrastructure, which is a critical enabler for all components of the Digital Decade policy programme.

The technological change around us is accelerating and offering new opportunities (artificial intelligence, 5/6G, new computing technologies (AI, quantum), bio-green-nano technologies, etc. The time of mere "digitization" is over for Estonia. Now the question is how to skilfully manage emerging technologies to bring about the broader societal changes that they enable. A

major challenge for organizations - both in the public and private sectors - is to keep up with the changes. This was shown by the crises - COVID and Russia's war in Ukraine. Understanding the situation and realizing the final goal are insufficient if the complexity of implementing the necessary changes exceeds the capacity of our companies and institutions. Therefore, a successful digital strategy must also offer a management model and understand how to simplify the complexity for the implementers.

The main goals of the whole of government strategy "Estonia 2035" and the currently valid development plan "Estonian digital society 2030" are the continuous movement towards an intelligent, caring, cooperation-oriented and innovative society. Based on this vision, the Estonian people of the future will be supported by a personal state, where the use of public services is as convenient as possible and considers the needs of the person, and people also have comprehensive control and an overview of the activities and data that concern them. Estonian people and companies are helped by the public sector, which is based on knowledge and data-based management logic and is open, transparent, and efficient in its activities. Estonian companies are smart and innovative, among other things, data and solutions based on artificial intelligence are widely used in the provision and development of their services.

This vision is broad and comprehensive, but the reality of the implementation of the development plan has been more modest - today's development plan deals more narrowly only with the digital state, cyber security, and connectivity.

Several activities of today's development plan, especially the most horizontal ones, have not yet been successfully put into operation, and are lacking a trajectory on how to get there: (green transition, innovation, cross-disciplinary cooperation, digital skills). These are topics whose responsibility and activities are shared between several development plans and programs.

In addition, the Government is leading a zero-budget design process, which will change the fundamentals of state budget programming. In connection with this, a broader change in state administration and supranational service-based management is underway.

Considering digital skills, the situation of education in Estonia has been described in the context of the three strategic goals of the Education Strategy, through strengths and challenges highlighted in studies and analyses, vision documents, education strategy working groups and stakeholder consultations. We are guided by the principle that to achieve the future goals of education in Estonia, it is necessary to maintain and further develop its strengths and to address bottlenecks.

In the digitalisation of businesses, the rapid change of technology is changing business models and the nature of work. To stay competitive, existing solutions need to be constantly updated and new ones developed. For enterprises, these changes create new business opportunities; however, exploiting them requires the ability (resources, skills) to adopt new technologies, the willingness to continuously learn and adapt business models, the ability to cooperate internationally and to find one's place in changing global value chains. This, in turn, requires keeping up to date with the latest developments in science and technology, both in the field of research and in the education system. The use of data, including data management and cyber

security, is becoming increasingly important, offering new business opportunities while also requiring increasingly thorough knowledge and skills.

CHALLENGES AND OBSTACLES THAT HINDER PROGRESS

Based on the changes described above, Estonia's aim is to bring together the internal and external developments that took place after the current development plan was drawn up (in 2020) under a personal state vision and a renewed structure, which would also consider the goals of the EU's Digital Decade. The new version must be the one document for the reader, where answers can be given to the question of what the vision of the Estonian digital society is and through the fulfilment of what goals, activities, metrics, and standards we are moving in this direction. What are the milestones from year to year and what investments will the implementation of these goals require.

The reason and obligation to spell out the metrics, trajectory, and investments in more detail stems from the objectives of the European Union's Digital Decade agreed in 2021. Together with the directions of the EU Digital Compass, they entail the obligation to draw up this national roadmap for the goals of the Digital Decade. These four directions are:

1. Digital skills: The area covered today by the Education strategy 2021–2035, where the national target of 60% (using a different metric system from the EU) is clearly below the EU target of 80%. The Commission's guideline for Estonia is that it should *speed up the efforts in the field of digital skills. It should strengthen measures to improve basic digital skills, e.g., encouraging employers to upgrade and retrain employees during working hours.*
2. Digital infrastructure: Here, Estonia is lagging the EU average the most, and the Commission's guideline for Estonia indicates, among other things, the direction of topics that we do not deal with according to today's development plan. *Estonia should speed up its efforts in the field of digital infrastructure. It should put in place measures to encourage the development of 5G connectivity and to incentivise the uptake of gigabit and 5G connectivity. Ongoing activities related to the quantum computer should be continued with increased coordination and cooperation to promote the quantum community across the EU.* In addition, we currently lack national targets for the development of future computing technology (*presence of edge nodes, cloud use of enterprises*).
3. Digitalisation of the public sector: In this area, Estonia has already fulfilled the general European 2030 KPI, but the Commission's guideline for Estonia is that *"Estonia should continue to implement the policy of digitising public services. Estonia should continue to invest in the procurement of innovation for digital solutions to further accelerate the adoption of innovative digital solutions in all public services. It should also address the availability of health data to enable access for all people (including persons with disabilities) and to ensure access to all relevant portals from mobile devices" (2023 Digital Decade Estonia report).*
4. Digitalisation of businesses – The Commission's guideline is *that Estonia should accelerate its efforts in the field of digitalisation of companies. Estonia should increase*

the uptake of advanced digital technologies in companies and support SMEs in using digital technologies to make them more competitive and sustainable.

STRENGTHS AND ASSETS TO BE LEVERAGED

Estonia has a vision and an action plan concerning the development of the Estonian economy, state, and society with the help of digital technology in the next decade. Regarding digital society, there persists an ambition to use information and communication technology, i.e., digital solutions as astutely and as much as possible to achieve the objectives of ‘Estonia 2035’ strategy. Namely, contributions are made via the following sub-objectives:

- The Estonian economy is innovative and knowledge-based, using new technologies and business models as well as flexible forms of work. The contribution of the development plan lies in the creation of favourable conditions for business research and development and innovation. The Estonian economic environment invites people to work here, establish companies or do business virtually, invest, create, and test new solutions which benefit the society at large.
- The needs of all people are considered when shaping the living environment and the foundations of high-quality spaces and principles of inclusive design are consistently followed when making decisions to ensure the accessibility and convenience of spiritual, physical, and digital space for everyone. The activities of the development plan contribute by innovative technologies and environmentally friendly solutions which reduce the impact of climate change and the time required for covering distances and ensure a good living environment across Estonia.
- Estonia is an innovative country which values the creation and use of knowledge and where social life is organised by means of new human-centric and efficient technologies. The contribution of the development plan is reflected in the fact that the governance set-up promotes social cohesion, the adoption of new solutions, innovation, and flexible governance. Public services function in the background and are proactive, and the data space is protected. The organisation of governance and people’s participation in it in Estonia is trendsetting and sets an example to other countries.

SECTION 2: ESTONIAN NATIONAL TRAJECTORIES AND TARGET VALUES TO CONTRIBUTE TO THE EU’S DIGITAL TARGETS

Based on the current strategy, Estonia’s the goal for the next decade is to increase Estonia’s digital power: digital government guarantees the best experience, high-speed Internet is available to all those who request it in Estonia and our cyberspace is safe and reliable. The goal is measured based on the development of four metrics as follows:

- Satisfaction of private persons with public digital services + satisfaction of entrepreneurs with public digital services + availability of highspeed Internet + resilience and trustworthiness of cyberspace /4=X
- Initial situation of the development plan in 2021 was $(69\% + 47\% + 58\% + 96\%)/4 = 67,5$. Objective for 2030 is $(90\% + 90\% + 100\% + 96\%)/4 = 94$

SECTION 3: POLICIES, MEASURES AND ACTIONS TO ACHIEVE THE DIGITAL TARGETS

OBJECTIVE I: Digital skills

Estonia scores just above the EU average on basic and above basic digital skills. In 2021, 56% of individuals had at least basic digital skills, which is slightly above the EU average of 54%. In 2021, 28% of individuals in Estonia had above basic digital skills which is two percentage points above the EU average of 26%. In 2021, the percentage of individuals with at least basic level skills in using software for digital content creation was 66%, the same as the EU average. Estonia is currently implementing several measures that can help increase the level of digital skills to contribute to the collective efforts to meet the Digital Decade target of at least 80% of the population having at least basic digital skills.

First, a programme was launched to empower public libraries to improve the digital skills of citizens living in their local area and enable them to use digital public services. The programme is supported by funding from the European Social Fund. As a first step, training is provided to librarians and municipality officials who will provide counselling services to local inhabitants with a low level of digital skills. Second, the Ministry of Education and Research launched an initiative to offer “in-service training courses” for adults. These courses are offered by vocational schools, universities, and other higher education institutions and are free of charge for the participants. Around a third of these courses focus on ICT skills. Third, training on digital skills is also funded by the Unemployment Insurance Fund. Around 10% of the total training budget is allocated to ICT training. To date, around 5 000 people have participated in these training courses.

According to the Country Report, the percentage of employed ICT specialists is higher than the EU average. With 6.6% of ICT specialists in the workforce in 2021, Estonia was two percentage points above the EU average. Given the high percentage of ICT graduates in recent years (10.1% in 2021 which is more than the double the EU average of 4.2%), the percentage of ICT specialists is likely to increase over the next seven years. This would further contribute to the Digital Decade target of having at least 20 million (around 10%) employed ICT specialists in the workforce within the EU by 2030. The share of female ICT specialists in Estonia, at 24.5% is also well above the EU average of 18.9% and among the highest in the EU.

Estonia has launched several programmes and initiatives to address the growing demand for ICT specialists² and ICT graduates in the labour market, and to promote women’s access to the field. First, recently launched retraining and upskilling programmes are expected to train another 7 000 ICT specialists between 2021 and 2027. These initiatives include three pilot programmes aimed at creating alternative paths into the ICT sector for people without previous training or work experience. Second, to reduce the gender imbalance in the ICT sector, the Estonian Ministry of Social Affairs has developed projects, supported by EU funding, to promote women’s access to the ICT sector and to reduce the current gender imbalance. This is in line with Digital Decade targets and objectives to reduce the digital divide and promote

convergence between genders. The design and implementation of these projects builds on the findings of an 18-month research project on “Glass Walls and Ceilings in the Estonian ICT Sector”. Third, an e-services development training programme will be piloted and launched in 2023. This programme is aimed at equipping service owners with the right skill set to develop new services and to re-think existing offerings, applying design thinking methods and a user-centric and problem-oriented approach. Estonia’s ambition to make a significant contribution to the collective efforts to achieve the 2030 Digital Decade targets for ICT specialists is also underlined by the fact that Tallinn University of Technology (TalTech) is a leading partner in the international consortium Digital4Business which aims to create a commercially sustainable 100% online European master’s study programme.

Enterprises increasingly provide ICT training and thereby contribute to improving the digital skills of their employees. The percentage of enterprises providing ICT training is with 19% still below the EU average of 22%. However, the gap between Estonia and the EU average is decreasing, from 8 percentage points in 2017 to 3 percentage points in 2022. This shows the positive impact of the measures taken by Estonia. The development between 2020 and 2022 was very similar to the EU average (which saw an increase of around 10%).

Learning options are responsive to the development needs of society and the labour market. Rapidly changing societies and evolving labour markets, including new fields and professions and new forms of work, require the upgrading of existing skills. This implies that individuals must be prepared for career transitions and for continuous learning – retraining and further training throughout their lives – and that the education systems are able to respond quickly to changing skills needs. By providing skills, the education system creates the preconditions for entrepreneurship and innovation, for economic growth that considers the specific characteristics of Estonia, and for a balanced and cohesive society. An ageing population will create new challenges for the labour market and the social sector, changing the way work is done and the expectations of future employees. Demand for technology-related skills will continue to grow until 2035. In a society based on new technologies, education plays an increasing role in the cultural and ecological survival of society. General and future competences, and at least a few in-depth competences in specific fields (the ‘T-shaped skills model’), help people to adapt in the labour market. Flexible and practical upskilling opportunities and work-based learning are becoming increasingly important in education.

Target for 2035:

IT education: the digital content development skills in all age groups create equal opportunities for all learners and conditions for increased competitiveness.

Current strategic goal indicator:

Digital competences above base level among 16 to 74-year-olds (%) from 37% in 2019 to 60% in 2035.

Trajectory:

Developing and implementing a sustainable system of forecasting and monitoring labour and skills needs which considers the needs of all target groups and of coordinating actions between different actors to promote the acquisition of knowledge and abilities that serve the labour market and to better link education to the labour market.

Essential actions:

- Developing a system for monitoring, forecasting, and providing feedback on skills needs for wider use of results.
- Creating a digital solution for managing individual learning pathways and careers, and for skills assessment (digital education history)

Trajectory:

Supporting the development of competences that create more added value; improving continuing training and retraining opportunities, including work-based learning, to respond swiftly to the development needs of the world of labour and ensure that people are equipped with the right knowledge and skills for employment.

Essential actions:

- Developing digital literacy across all age groups to increase digital inclusion and develop skills in creating information technologies.

OBJECTIVE II – DIGITAL INFRASTRUCTURES

According to the Country Report, there is room for improvement regarding Estonia's contribution to the Digital Decade targets on connectivity. Estonia's 5G coverage and the take-up of fixed broadband of at least 100 Mbps are low. There is a clear need to act to be able to contribute to the collective efforts to achieve the Digital Decade connectivity targets.

Estonia is in the upper half of Member States when it comes to fixed very high-capacity network (VHCN) coverage. 79% of households are covered by a fixed VHCN, which is above the EU average of 73%. The increase of 6 percentage points compared to the 2021 figures is also above the EU average. Looking at the development over the last five years, the gap between the EU average and Estonia is even increasing. Coverage of fibre to premises (FTTP) is in the top third of Member States. 79% of households covered is significantly above the EU average of 56%. The growth rate from 2021 to 2022 is, however, slightly below the EU average. Significant efforts are needed to meet the Digital Decade goal of ensuring all end users at a fixed location are covered by a gigabit network up to the network termination point.

According to the report, Estonia is in the bottom five Member States for take-up of fixed broadband of at least 100 Mbps. With 29% of households subscribed to fixed broadband of at least 100 Mbps, Estonia is significantly below the EU average of 55%. Since 2021, the at least

100 Mbps take-up of fixed broadband take-up has increased by seven percentage points, which is above the average growth rate in the EU. This is mainly due to private investments to upgrade old copper networks and public investments in rural areas. However, over the last five years, the gap between Estonia's scores and the EU average has increased. The take-up of fixed broadband of at least 1 Gbps remains at 0% of households. This is significantly below the (fast growing) EU average of 14%. The take-up of mobile broadband was the same as the EU average of 87% in 2021.

Further efforts are needed to contribute to the Digital Decade target of having full coverage of all populated areas by 2030. Overall, 5G coverage in Estonia has reached 43% of populated areas, which is significantly below the EU average of 81%. Regarding 5G coverage on the 3.4-3.8 spectrum band, which is necessary for advanced applications that need large spectrum bandwidth, Estonia has only achieved 15%. This is again significantly below the 41% in the EU overall. Recent improvements in 5G spectrum assignment will improve the 5G coverage over the coming years.

Estonia has taken several measures to improve its connectivity infrastructure to contribute to the collective efforts to achieve the Digital Decade targets in this area. These include support measures to finance the construction of VHNC networks in rural areas. Around EUR 24 million from the RRF and EUR 45 million from the ERDF have been allocated to build VHCN in rural areas between 2023 and 2027. Regarding 5G, based on the most recent information provided by Estonia, the 700 MHz, 3400-3800 MHz, and 26 GHz bands have been assigned. Mobile operators are continuing to roll out 5G networks.

Estonia has no developments to report in semiconductors and edge nodes. Estonia contributes to the collective efforts in high-performance computing mainly through the Estonian Scientific Computing Infrastructure (*ETAIS*) project. The project is run by a consortium of the University of Tartu, Tallinn University of Technology, the National Institute of Chemical Physics and Biophysics, and The Education and Youth Board (HARNO). ETAIS aims to increase the competitiveness of Estonia's computing and data-intensive research disciplines by providing access to a new and modern scientific computing infrastructure. Estonia is also part of the EuroQCI initiative to build a pan-European quantum infrastructure and a partner in the Nordic-Estonian Quantum Computing e-Infrastructure Quest (NordIQuEst). The purpose of this project (which runs from April 2022 until May 2025) is to create a Nordic ecosystem that combines high-performance computing and quantum computing.

Target based on the general objective:

By 2030, ultrafast, reliable, and affordable telecommunications connections are available in Estonia irrespective of your location, making it possible to create and use innovative services.

This sub-objective is directly related to the implementation of one of the axes of the vision of digital society: creating an impressive way of life. As has been stated above, the vision sets out that services should be convenient and available all over Estonia, which is why relevant

(telecommunications) connections are needed. The digital power of the economy, in turn, depends on the development of connectivity. The better the connections, the more digital solutions, products, and services can serve as vehicles for development in various sectors and across Estonia. The quality and availability of connections also determines whether conditions are favourable for the creation of future solutions in Estonia and whether people from other countries come here with the same aspirations.

The metric used to measure this target: The proportion of Estonian households and companies who can access an Internet connection of at least 100 Mbit/s which can be increased up to 1 Gbit/s. The objective is to move from 58% in 2021 to 100% in 2030.

OBJECTIVE III - DIGITALISATION OF BUSINESSES

Estonia acknowledges a mixed picture regarding the uptake of digital technologies by enterprises: In 2021, 51% of enterprises in Estonia used cloud services, which is significantly above the EU average of 34%. At the same time, only 10% used big data in 2020, and even fewer (3%) used AI in 2021. Both figures are below the EU average. The total percentage of enterprises using one of the three technologies remains below the Digital Decade target of at least 75%. Only 3% of enterprises were reported to be using AI technology in 2021, which is significantly below the EU average of 8%.

Several measures are taken to increase the percentage of enterprises using big data, AI, and technology more broadly. For example, an AI development programme has been created focusing on industrial companies and start-ups in the areas of green, health, and energy. The ‘Digitalisation Roadmap’ programme is intended to provide funding for SMEs for dedicated three-year action plans and services to improve their digitisation. Estonian enterprises can also apply for support for automation and the deployment of digital technologies and robots under a programme using RRF resources. The latest addition (March 2023) is a programme providing support to SMEs for cyber security measures, based on methodology developed by Estonia’s Information System Authority (responsible for applying information security standards in the public sector).

The Estonian Digital Innovation Hub ‘AI and Robotics Estonia’ (*AIRE*)⁵ also focuses on improving the AI capabilities of Estonian companies and developing their digital maturity. It is part of the network of European Digital Innovation Hubs (EDIH). The EDIHs are one-stop shops supporting companies and the public administration to respond to digital challenges and to become more competitive. The EDIH network is co-financed by the European Commission and by the relevant Member States of the EU, in this case Estonia. *AIRE* in Estonia focuses mainly on the manufacturing sector (90% of the hub’s work), but some resources are allocated to the health and tourism sectors. *AIRE* has been fully operational since 1 July 2022, after a 12-month pre-launch testing and pilot phase.

Estonian enterprises also present a mixed picture on the uptake of other digital technologies. Electronic information sharing in enterprises is low compared to other Member States. Only

23% of enterprises reported using an ERP (enterprise resource planning) software package to share information between different functional areas. This is significantly below the EU average of 38%. The use of social media is also below the EU average. Only 22% of enterprises reported to use two or more of the social media tools considered as part of the assessment. This is again significantly below the EU average of 29% of enterprises. By contrast, the uptake of e-invoicing for automated processing is very high in Estonia and has almost doubled between 2018 and 2020. 62% of enterprises in Estonia reported to be sending e-invoices.

67% of SMEs had at least a basic level of digital intensity in 2022. The percentage of SMEs in Estonia selling online is 19%, the same as the EU average. The growth rate since 2017 has also been like that of the EU average. Estonian SMEs' total turnover from e-commerce is 12%, slightly above the EU average of 11%. The percentage of SMEs in Estonia carrying out electronic sales to other Member States is 9%, the same as the EU average.

The Estonian business ecosystem includes many innovative and growing start-ups and scale-ups that are driving the country's growth and modernisation. There is currently 10¹ unicorns born in Estonia or founded by Estonians².

Digital solutions across all areas of life

In the current situation, where Digital Decade targets are divided between various Estonian development plans, the Estonian "R&D, innovation and Business Development Plan 2021 - 2035" sets goals to reach the objectives above.

With the support of R&D, innovation and entrepreneurship, digital solutions are created, delivered, and used in every area of life. Data economy is used to create new business opportunities, secure cyberspace is ensured.

Using Information and Communication Technology (ICT) horizontally through other sectors was selected as a growth area during RDI strategy 2014-2020 preparation. The development of the ICT sector continues to be important, as it provides a major local (e-governance) and global (export of smart solutions and e-services, cyber security, application of artificial intelligence, etc.) contribution to the functioning of society and the economy.

In recent years, R&D has mainly focused on:

- Artificial intelligence and machine learning - innovative solutions for creating automated systems, including healthcare (e.g., ICT-based solutions for e-health and personalised medicine) and e-government solutions.
- Data science and big data – solutions for user-centred e-Governance, including the Internet of Things, robotics, industrial automation, healthcare, etc.

¹ [The full list of Estonian unicorns — Invest in Estonia](#)

²² [Unicorns list | Estonia Innovation Ecosystem \(startupestonia.ee\)](#)

- Robot-human interaction and the Internet of Things in industry, including the digitalisation of industry and optimisation of processes.
- Reliability of software, including measures of software correctness, safety, resource usage analysis, validation of analytical results and methods and tools for independent verification of certificates (e.g., for e-government and public services).
- The Internet of Smart Things, including both devices and technical solutions, as well as changes in processes (e.g., production processes) and in people's daily behaviour (e.g. health data).
- Security and reliability of hardware and systems, cyber defence – solving security problems related to cyber threats (e.g., in the fields of energy, communications, finance, transport, and security).
- Digital transition and lifelong learning by addressing not only the information technology dimension but also the social, cultural and values dimensions in the development and use of digital solutions for lifelong learning; the assessment and development of digital maturity of organisations and the adoption of digital innovations in education and entrepreneurship.

The development plan is linked with underlying strategic guidance from the “Estonia 2035” strategy, in particular the business environment, the cultural space and the living environment, security and safety, public administration.

On EU level, the strategy considers: Objectives of EU digital transition, IPCEI value chains United Nations Sustainable Development Goals (SDGs): those on employment and growth, industry, innovation and infrastructure, peaceful and inclusive societies, and global cooperation.

2. Estonia's strengths and development opportunities Economic dimension: The role of the ICT sector in the Estonian economy is significant¹ – with around 5% of the total number of people employed in enterprises, it generates nearly 8% of the value added created by enterprises, pays over 6% of all taxes and levies withheld and paid by Estonian enterprises, and produces 11% of our exports. At the national level, it is important to develop and export e-services to other countries, which puts Estonia in a unique position as a world leader in this field. The Smart Specialisation Growth Areas Survey (2019) tracked the progress of the growth area ‘ICT horizontally across other sectors’ by niches and showed above-average economic growth compared to all economic activities. Scientific dimension: most of the R&D in ICT is concentrated within the University of Tartu and TalTech (which are also responsible for the provision of higher education in ICT), in addition to Tallinn University, STACC OÜ, and Cybernetica AS. TalTech has the highest number of ICT researchers (more than 200 jobs), while the University of Tartu has around 150 ICT researcher jobs. Each year, researchers in the field of ICT publish more than 200 research publications, of which around one in six are among the top 10% of the world's most influential publications.

BUSINESS ENVIRONMENT: Estonia's business environment is conducive to the entrepreneurial spirit and to the emergence and growth of knowledge-intensive enterprises, the

creation and export of higher value-added products and services, and investments in all regions of Estonia Estonia's position in the Doing Business index Assessment of the competitiveness of the Estonian business environment Position in the ranking 18 2020 Estonia in TOP 5 Export of goods and services Assessment of the international competitiveness of the Estonian enterprises Billion euros 19 2020 43 GDP per capita generated outside Harju County of the EU 27 average Regional dimension % 41 2020 59

Digitalisation of businesses

1) Activities of strand 1 of the Business Environment "Developing a competitive and smart business and consumption environment" 1.4. raise awareness among entrepreneurs about the so-called twin transitions (digital and green transitions for a sustainable and digital economy) and business transfer, and provide support services to foster technological innovation, the use of the best available technologies, the rise in management quality and generations exchange, p. 19.

(2) Activities of strand 2 of the Enterprise Environment "Increasing the capacity to create higher added value and export" 2.1. support the use of the best available technologies in industrial enterprises and encourage the adoption of business models based on modern technologies, allowing, inter alia, the diagnosis and audit of bottlenecks and the involvement of experts and top specialists from abroad with international experience, p. 20.

3) Activities of the knowledge transfer strand 1 "Increasing the R&D&I intensity and knowledge transfer capacities of enterprises" 1.9. encourage the automation of enterprises through the development of a digitalisation roadmap and the financing of investments therein, including support for digitalisation and the use of AI and robotics-related technologies to make companies' processes and supply chains more efficient and to increase the added value of products and services, p. 16. 4) Knowledge transfer 1. Under the strand, four areas of smart specialisation have been selected, which the state is prioritising by directing additional resources there. One of the priority areas of development is "Digital solutions in every area of life", p. 13.

Innovative companies/scale-ups (unicorns)

The "R&D, Innovation and Business Development Plan 2021 - 2035" aims to meet Digital Decade targets by focusing on innovative companies and scale-ups, notably unicorns. The plan includes various activities implemented through a knowledge transfer program in R&D and entrepreneurship. Key actions involve enhancing R&D&I intensity and knowledge transfer capabilities among companies and fostering science and technology-intensive start-ups. To develop the start-up ecosystem and improve capital access, several initiatives are underway. These include support conditions for boosting start-up entrepreneurship through Startup Estonia, which focuses on ecosystem development, talent, and capital via accelerators. Investments from the Baltic Innovation Fund (BIF), EstFund, and AS SmartCap in start-ups and venture capital funds are critical for developing the local capital market and enhancing

capital availability. Support is also extended for deep tech start-up projects, the Creative Destruction Lab-Eesti (CDL-Eesti) business accelerator, the NATO DIANA business accelerator, Applied Research Programme Grants (RUP), and Applied Research Centre Services (RUK).

OBJECTIVE I - DIGITALISATION OF PUBLIC SERVICES

Estonia is at the forefront of the digitalisation of public services and is expected to make a very strong contribution to the collective efforts to achieve the Digital Decade targets for the digitalisation of public services.

The country can serve as an example to other Member States. Estonia has shared its experience and best practices with other countries, including Ukraine as a candidate country, as part of the Estonian e-Governance Academy, a foundation-based international development cooperation programme. Further advancements in digital public services remain one of the priorities under Estonia's national Digital Agenda 2030.

Estonian citizen and businesses can access a wide range of public services online: With a score of 94, Estonia is one of the very advanced Member States in terms of the proportion of administrative steps that people can carry out online for their major life events (the EU average score is 77). Estonia's score of 99 is significantly above the EU average score of 84. This indicator broadly reflects the share of public services needed to start and run a business that are available online for domestic and foreign users. Estonia's score for data that is pre-filled in online forms for public services is 88, which is also significantly above the EU average of 68. Internet users in Estonia make good use of the user-friendly services provided: 93% of internet users in Estonia used the internet to interact with public authorities in the last twelve months. This is significantly above the EU average of 74%. Online public services in Estonia are generally considered to be particularly transparent and designed with user involvement. Users have a high degree of control over how they manage their personal data. With a score of 87, Estonia is significantly above the EU average score of 65. In terms of user support Estonia is also one of the highest-ranking countries. The score of 98 indicates excellent online support, help features, and feedback mechanisms. The EU average score is 84. User support was at a level well above the EU average when it was first measured in 2021 (with a score of 93 compared with an EU average of 82).

Estonia is committed to further improving and developing its online public services, as one of the three priorities set out in the national Digital Agenda 2030. The focus is currently on linking services to life events and developing features that automatically suggest further steps that are typically linked to other services or a particular life event (e.g., marriage may lead to changing the surname which triggers the need to update several documents). Another priority is providing the user interface and allowing for a seamless interaction with public services, irrespective of when and where, or from which device, the services are being accessed. With a score of 97, the 'mobile friendliness' of online public services in Estonia is already above the EU average of 93, and Estonia is committed to improving it further. To further strengthen e-governance, Estonia created a centre of excellence for data governance and open data. This centre improves

the quality of the data collected and held by the Estonian public authorities, increasing its usability for decision-making and its availability as open data to other stakeholders.

Estonia is at the forefront of providing access to electronic health records, and currently ranks 4th among the Member States. 89% of Estonians have access to their electronic health records, which is significantly above the EU average of 72%. This is a significant contribution to the collective efforts to reach the Digital Decade target of 100% access. Citizens have full access to all their electronic health data (including medical images in digital format) through a national online access service (using eID authentication). A mobile application is not available. The data is updated after each healthcare encounter. Real-time remote assistance is available in case of problems accessing the data. The service is offered to the whole population and supplied with relevant health data from the entire care sector, including both public and private providers.

Accessibility can be improved. The electronic access service in Estonia is not compliant with the Web Content Accessibility Guidelines v2.1 (WCAG), as required by the European Commission's Web Accessibility Directive. Estonia reported that monitoring at national level shows that digital health records are largely accessible, but older people and people with disabilities may experience some difficulties. These difficulties are expected to be addressed when a new health portal will be rolled out.

Most of the Estonian population has access to an eID scheme notified under the eIDAS Regulation which provides access to a wide range of digital public services. Currently, Estonia has six eID means notified under the Estonian eID scheme. The ID card, the RP card, the Digi-ID, the e-Residency, the Digi-ID, the Mobiil-ID, and the Diplomatic identity card have all been notified at a 'high' level of assurance. Estonia therefore makes a significant contribution to the collective efforts within the EU to achieve this target. Estonia has not reported any specific actions taken at national level to further increase the percentage of citizens with access to an eID scheme.

Target based on the general objective:

to ensure the best digital government experience by 2030.

This sub-objective is directly related to the implementation of one of the axes of the vision of digital society 2030: creating an impressive way of life in Estonia by means of digital solutions. While the previous aim set for the development of digital government was to primarily increase the efficiency of public administration, the quality and human-centricity of services is the next level of maturity in digital government. This does not mean that things should not or cannot still be done more efficiently – it must be done in addition to and while improving the quality of services. In addition, the implementation of this objective contributes to the progress of other axes of the vision. Public services that provide the best experience and are of high-quality help to make the economy more digitally powered because they simplify procedures for all entrepreneurs – among other things, it increases the number of e-residents. To ensure the best experience of digital government, steps must be taken to establish favourable conditions for the creation of future solutions that improve our experience, as has been highlighted in the vision.

The metric used to measure this target:

Satisfaction with public digital services among private persons from 69% in 2019, with an 90% objective for 2030. Among entrepreneurs, from 47% in 2019 to 90% in 2030. This metric measures the satisfaction of private persons and companies with public services which have been used via electronic channels (electronic self-service, websites, e-mails, applications, social media). The result is sent as an extract from the catalogue of public services, and it is measured once a year. The lines of action required for achieving the subobjective are divided in two because we are tackling two challenges at once: taking digital government to the next stage of development by means of new solutions and keeping the established solutions sustainable and updated.

BUDGET ALLOCATED OR PLANNED

Estonia's Digital Agenda 2030

The estimated cost includes the total assessment of the costs MEUR (without VAT) that are required for ensuring sufficient funding for the achievement of the objectives of the Digital Agenda via the digital society programme:

Year	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	Total 2021- 2030
Cost (million €)	68	111	135	120	120	125	131	135	138	139	1224

The detailed programme with allocated budget can be found at:

<https://www.fin.ee/media/7665/download>

Education strategy 2021–2035

Estimated cost of the strategy

Year	Description of costs			Total (EUR millions)
	Baseline and Strategic Goal 1	Strategic Goal 2	Strategic Goal 3	
2021	882	65	29	976
2022	976	87	60	1122
2023	1011	153	58	1222
2024	1045	220	64	1330
2025	1099	291	68	1457
2026	1153	331	71	1556
2027	1206	371	72	1649
2028	1221	410	70	1701
2029	1264	445	62	1771
2030	1280	481	45	1807
2031	1342	529	46	1917
2032	1407	580	46	2033
2033	1474	633	47	2153
2034	1544	688	47	2280
2035	1618	747	48	2412

The forecast shows the expected cost in millions of euros. The forecast is based on the population projection of Statistics Estonia and the long-term economic forecast of the Ministry of Finance. Baseline and Strategic Goal 1: Costs of providing formal education, including operating costs, subsidies and investments; means to maintain teachers' salaries at the 2020 reference level; continued restructuring of the school network in line with the principles of spatial quality. Strategic Goal 2: Costs to increase adaptability and flexibility in education (learners with special educational needs, learners from migrant backgrounds); development of teaching resources and curricula and teacher training costs; increasing teachers' salaries to 120% of the national average by 2025 and maintaining this level until the end of the period. Strategic Goal 3 includes investment in the activities of Action Trajectory 3, i.e. vocational systems, OSKA, adult learning, apprenticeships, work-based learning, etc.

The detailed programme with allocated budget can be found at:

<https://www.hm.ee/media/3808/download>

The most recent budget allocations for the Estonian digital agenda:

Program activity - name	in thousand euros				
	2023 budget	2024 budget	2025 RES	2026 RES	2027 RES
Leaps of development of the digital state	-3 647	-3 025	-3 123	-1 877	-1 866
Securing the basic base of the digital state	-91 208	-95 895	-100 459	-99 680	-102 747
				-101	
	-94 855	-98 920	-103 581	558	-104 613
Management and coordination of national cyber security	-877	-1 526	-1 041	-1 045	-1 043

Developing the ability to analyse trends, risks, and impacts	-427	-327	-330	-336	-334
Ensuring cyber security	-9 010	-14 241	-13 681	-12 673	-12 676
	-10 314	-16 093	-15 053	-14 054	-14 052
Ensuring legal space	-10 036	-3 105	-3 106	-2 973	-2 866
Development of access networks	-10 161	-25 226	-15 317	-15 166	-163
Development of 5G infrastructure and services	-169	-6 881	-4 162	-4 166	-4 164
Total measure	-20 367	-35 212	-22 585	-22 305	-7 193
				-137	
Total program costs	-125 536	-150 225	-141 219	917	-125 858

Estonian Research and Development, Innovation and Entrepreneurship Strategy 2021—2035

RDIE funding from 2021 onwards 1% of GDP*

Research funding will reach 1% of GDP by 2021 and will be maintained at least at the same level thereafter, subject to public finance policy possibilities. The 1% of GDP projected in the Strategy's budget forecast does not mean that the level of research funding could not rise above that level.

RDIE Strategy	Funding 2021-2035 (million EUR)
Research System Programme (MoER)	2,828
Knowledge Transfer Programme (MoER+MoEAC)	3,361, incl. MoER 2024 + MoEAC 1,337
Business Environment Programme (MoEAC)	1,708
Total funding for RDIE Strategy	7,897

* In addition to the funding scheme of the RDIE Strategy, sectoral R&D activities are funded by other ministries under their own strategies. The sectoral R&D spending of other ministries is not included in the funding scenarios of the RDIE Strategy.

The assumption is that the nominal funding of the research system will be maintained even during an economic downturn.

The detailed programme with allocated budget can be found at:

<https://www.taie.ee/media/434/download>

SECTION 5:EU LEVEL COOPERATION

ESTONIAN PARTICIPATION IN MULTI-COUNTRY PROJECTS AND JOINT COMMITMENTS

OVERVIEW OF EXISTING AND POSSIBLE FUTURE MCPS AND JOINT COMMITMENTS

Estonia is collaborating with other Member States in setting up two **European Digital Infrastructure Consortia (EDICs)** in digital public services. The Networked Local Digital Twins Towards CitiVerse is intended to help build carbon neutral, resilient, inclusive, sustainable, and more beautiful cities that are closer to people. The Genome EDIC aims to provide an infrastructure supporting genome-based medicine, research and innovation, and a better healthcare system for personalised medicine. Estonia is also considering the possibility of setting up the Copyright Infrastructure EDIC to define and promote the rules governing data management practices in creative industries. Estonia's national digital strategy already encourages the public sector to use innovation procurement as a strategic tool to accelerate the uptake of innovative digital solutions in public services. The review of Estonia's Digital Agenda 2030 will implement the recommendation of a national roadmap setting out ambitious initiatives in the digital sector, which will help to reinforce the country's investments and contribute significantly to the target on digitalisation of public services and the Digital Decade objective on innovation.

Overview 1) a. – multi-country projects included in the list of areas of activity for MCP in the Annex of the Decision, to which Estonia is committing or plans to commit in the future:

Name of the MCP	Short description of the MCP
Networked Local Digital Twins Towards CitiVerse EDIC	The digital twin for cities EDIC will help to build carbon neutral, resilient, inclusive, and sustainable cities more beautiful and closer to citizens. It will gather existing infrastructure solutions developed by individual Member States or in the context of the 'Digital Programme for cities and local communities', such as the EU Toolbox for Local Digital Twins. This EDIC will provide a long-term sustainable place for such solutions as well as improve data use, bring standardisation efforts for digital twins together and ensure interoperability between the digital twins from the start. It will lay down a foundation for the EU CITIVERSE and pursue joint activities together (e.g., joint public procurement or shaping the supply market). This EDIC has been pre-notified in June 2023 by EE, DE, SI, ES, CZ (as members). The proposed seat for this EDIC is Estonia.
Genome EDIC	The 1+MG multi-country project will improve the cooperation of MS in the

	achievements of the general objectives of the Digital decade through the formation of a European common data infrastructure and provision of services for data reuse in research and development for better healthcare. The aim is to provide an infrastructure supporting genome-based medicine, research and innovation and a better healthcare system for personalised medicine. This EDIC has been pre-notified in June 2023 by LU, FI, EE (as members) and support is still being collected from the Member States who had expressed their interest. The proposed seat for this EDIC is Luxembourg.
Alliance for Language Technologies (ALT-EDIC)	In a market dominated by international stakeholders, and where EU's data is not yet used to its full potential, the ALT-EDIC will establish a European-wide infrastructure and technology centre in the field of natural language processing. It will develop Large Language and Foundation Models optimising the uptake of European language technologies including through generative AI. The ALT-EDIC will address the shortage of training data in European languages and contribute to digital sovereignty, promotion of EU's values and linguistic diversity. The ALT-EDIC EDIC has been pre-notified in June 2023 by FR, LV, IE, SI, PL, HU, PT, ES, LT, EL, NL (as members), EE, AT, BE, BG, MT, RO (as observers). The proposed seat for this EDIC is France.

SECTION 6: STAKEHOLDERS INVOLVEMENT

The following is a non-conclusive list of actors in the engagement process for the ongoing renewal of the Estonia's Digital Agenda 2030:

Ministry of Economic Affairs and Communications as the central ministry for overseeing Estonian Digital Transformation, all other ministries, IT houses, State Information System Authority (RIA), Estonian Information and Communication Technology Centre (RIT), State

Info communication Foundation (RIKS), State Agencies (Consumer Protection and Technical Regulatory Authority (TTJA), Transport Administration, Estonian Academy of Music and Theatre (EAMT), Education and Youth Board (HARNO), Environmental Board, etc.), The Parliament, State Chancellery, Office of the President, Office of the Chancellor of Justice, Human Rights Centre, Development Monitoring Centre, Association of Estonian Cities and Rural Municipalities (ELVL), Estonian Association of Information Technology and Telecommunications (ITL), E-Estonia Centre Advisory Board, Enterprise and Innovation Foundation, Chamber of Commerce and Industry, Chamber of Service Economy, Tallinn University of Technology (TalTech), University of Tartu, Tallinn University, other training institutions, Nordic Institute for Interoperability Solutions (NIIS), AI & Robotics Estonia (AIRE), Merchants' Association, etc.

Currently, the first round of engagements is ongoing, where during 1-1 meetings the main partners will be engaged to define the bottlenecks of the development plan, renewal goals, principles, areas. The aim of the first phase is to map the concerns of the main partners and stakeholders regarding the current development plan. Based on this, the plan, topics and questions of focus groups and interviews will be derived for phase two.

The currently valid strategic documents, Estonian Digital Agenda 2030 (the national strategy for Estonian digital transformation), Education strategy 2021–2035 and the Estonian Research and Development, Innovation and Entrepreneurship Strategy 2021—2035, are based on a wide and inclusive programme of consultations with relevant stakeholders, according to national regulations.

The complete list of suggestions for the Education strategy can be found at: <https://www.hm.ee/media/1594/download>

The complete list for the Estonian Research and Development, Innovation and Entrepreneurship Strategy can be found at: <https://taie.ee/media/413/download>

The Complete list for Estonian Digital Agenda can be found at: <https://mkm.ee/digiriik-ja-uhenduvus/digihiskonna-arengukava-2030>

SECTION 7: OVERALL IMPACT AND CONCLUSION

To achieve the 2030 vision for a digital society, it is necessary that in other policy areas the local development plan is complemented in several ways. According to the organisation of the preparation of Estonian development plans, this is not an automatic obligation to other areas, but an order and expectation of the digital society vision drawn up within the framework of this development plan regarding the content and direction of other policies. They provide the basis for presenting views on the implementation and renewal of development plans and for coordinating activities between institutions.

Achieving the 2030 targets for the digital society will require:

The availability of sufficient and high-level ICT specialists, including the availability of cyber specialists, is critically important. By 2030, the number of ICT specialists in the economy needs to at least double, and the proportion of cyber specialists among them will have to increase to realise future goals to the extent necessary. To do this, the relevant study opportunities must be constantly expanded and the quality of study at all levels constantly improved – this requires, above all, decisive steps in the volume and quality of teaching capabilities, including ensuring the succession of teaching staff. Since the domestic education system alone is not enough to realise opportunities, it is necessary to create an easy opportunity for talents to come to Estonia – to make it a convenient and attractive destination for them.

To bring the wider digital transformation of the economy and public administration to the next level of digital maturity, it is important to initiate large-scale retraining and upskilling of digital skills. This will enable professionals from different fields to acquire the necessary digital expertise to launch and manage or support digital transformation in their organisations. The development of (professional) digital skills must also be a natural part of all levels of education. These steps will also facilitate workers' adaptation to changes in the economy.

The need to address the wider digital literacy of the population will continue. There is less and less need to "bring people online". Instead, it must be ensured that they have up-to-date skills to handle digital solutions usefully and safely. By 2030, all Estonian adults should be regular Internet users. This will provide an opportunity to ensure that they have sufficient capacity, including a basic level of relevant awareness, to be able to make ever better use of the services following the leaps in development of the digital state.

In the field of research and development, it is necessary to invest in the growth of research and development capabilities related to the development of the digital society. Then you will find players to create smart solutions and at the same time have the knowledge and solutions to try and implement quickly in the country and in the economy. The key is to ensure sufficient ICT funding for the Technical Regulatory Authority. The best possible services need a good Estonian language technology base so that both domestic and global service providers can make their services as convenient as possible for the members of the Estonian digital society.

This means increasing investment in basic language technology solutions.

Enterprise policy must focus on two objectives at once: (1) supporting the digital transformation of more traditional sectors (e.g., industry) with the necessary levers – from raising awareness and skills to supporting investment; 2) continuous development of a technology-based business environment. Excellent conditions must be created and ensured in Estonia, that companies develop smart products and services, that companies that create new smart solutions are born and grown, and that corresponding foreign players come to Estonia.

The development of the business environment also plays an important role in this, so that doing business in Estonia would be the easiest in the whole world. The development of so-called real-time economic organisation and solutions will play an important role as the next leap in development.

To ensure both the development of the business environment and, more broadly, the creation and deployment of innovative solutions, in a rapidly changing world, law-making must be flexible and able to respond quickly to opportunities. At the same time, law-making must continue to protect fundamental human rights and ensure ethical use of data.

If public services in a country are increasingly invisible, i.e., more proactive, and automatic, the interaction between people and the state and people's understanding of the functioning of the state may decrease. To avoid this, the quality of information services and the openness of governance must be improved. In other words, people must be increasingly actively involved in public decision-making and debate. This can be done, among other things, by making smart use of digital solutions, from supporting the functioning of communities to increasing opportunities for participation at national level.

Through business diplomacy and export support activities, it is necessary to continue working for Estonia to be known in the world as an ambitious and smart digital society and country. This opens doors for companies to bring IT solutions into the world, which makes them a stronger partner in creating future solutions in Estonia as well. On the other hand, Estonia's reputation depends on whether talents and world players consider Estonia as their possible next location.

Achieving the sustainability of the health and care system highlighted in the vision through digital solutions will only be possible if the health and well-being development plans focus on a comprehensive and meaningful digital transformation strategy, the development of content areas is well integrated, and the implementation is also effectively managed. This is demonstrated by the experience so far in managing the public health development plan and the lack of implementation of the e-health strategy.